Physical AI SDK - Model Lifecycle & Strategy 2026

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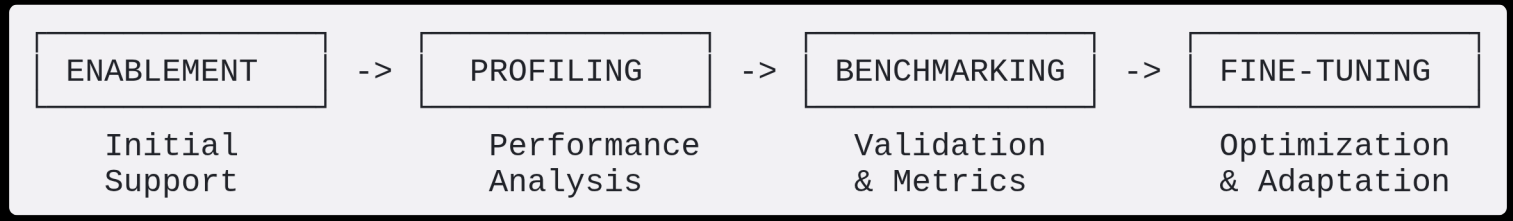
Model Lifecycle Overview



Key Phases:

1. **Enablement** - Initial model support,Insatallation, Off-the-shelf installtion, benchmarking(respoective metrics); Identify the right dataset, compare with right leaderboard of opensoure results.
2. **Profiling** - Performance analysis, bottleneck identification, W.R.T: latecny, compute efficiency, Can be generic, Step to identify what operation are more time consuming or compute consuming,
3. **Benchmarking** - performance metrics, competitive analysis, With a right dataset on gpu vs cpu npu
4. **Fine-tuning** - Model optimization writ. target environemnt(dirver, rocm, gpu),

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Progressive Model Support Strategy

From Individual Models to System Orchestration

Phase 1: Individual Model Support

- Focus on single model enablement and optimization
- Establish baseline performance for each model
- Build reusable infrastructure and tooling
- Might get simple as we gain expertise

Phase 2: Multi-Model Orchestration

- Systeems: Orchestration of multiple models working together (More complex)

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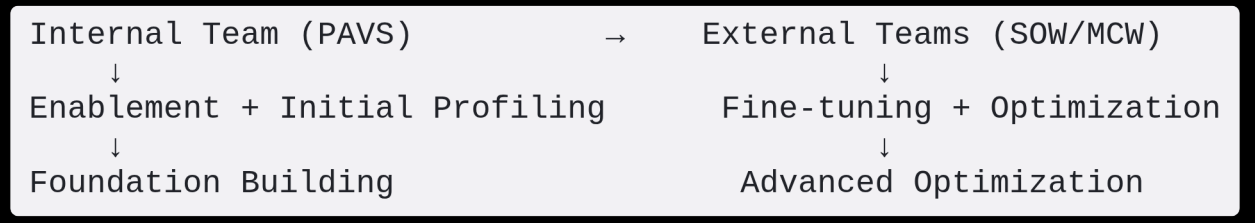
Effective Team Collaboration

Progressive Engagement Model



Strategy:

- **PAVS Team:** Handle enablement and initial profiling
  - Establish working baseline
  - Identify optimization opportunities
  - Create clear specifications for external teams
- **External Partners:** Focus on fine-tuning and advanced optimization
  - Domain-specific optimization
  - Large-scale benchmarking
  - Production deployment preparation

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Speed Matters - Parallel Execution

Model Status Across Lifecycle Stages

Model Name	Category	Enablement	Profiling	Benchmarking	Fine-tuning
MobileSAM	Vision/Segmentation	✅	✅	✅	🔄
YoloV12	Object Detection	✅	✅	✅	⏸️
Qwen3-VL	VLM	✅	✅	🔄	⏸️
LLaMA3.2-Vision	VLM	✅	✅	⏸️	⏸️
Whisper	ASR	✅	✅	✅	✅
XTTS	TTS	✅	✅	✅	⏸️
DeepSeek-R1	LLM	✅	🔄	⏸️	⏸️
Qwen2.5-VL	VLM	✅	✅	✅	🔄
CrossFormer	Vision Transformer	✅	🔄	⏸️	⏸️
CenterPoint	3D Detection	🔄	⏸️	⏸️	⏸️
OpenVLA	Vision-Language-Action	🔄	⏸️	⏸️	⏸️
MedSigLIP	Medical Classification	🔄	⏸️	⏸️	⏸️
EfficientNetV2	Image Classification	⏸️	⏸️	⏸️	⏸️
RAFT Stereo	Stereo Matching	⏸️	⏸️	⏸️	⏸️

Legend: ✅ Complete | 🔄 In Progress | ⏸️ Not Started

Key Insight: Start early, execute in parallel across multiple models

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Slide 6: Environment & CI/CD Infrastructure

Isolated Environments for Each Model/System

Environment Strategy:

```
physical-ai-sdk/
├── models/
│   ├── mobilesam/
│   │   ├── gpu/
│   │   │   ├── Dockerfile
│   │   │   ├── pyproject.toml (uv)
│   │   │   └── .venv/
│   │   └── npu_gpu/
│   │       ├── Dockerfile
│   │       ├── pyproject.toml (uv)
│   │       └── .venv/
│   ├── yolov12/
│   │   ├── gpu/
│   │   │   ├── Dockerfile
│   │   │   ├── requirements.txt (pip)
│   │   │   └── .venv/
│   │   └── npu_gpu/
│   │       ├── Dockerfile
│   │       ├── requirements.txt (pip)
│   │       └── .venv/
│   └── llama3-vision/
│       ├── gpu/
│       │   ├── Dockerfile
│       │   ├── pyproject.toml (uv)
│       │   └── .venv/
│       └── npu_gpu/
│           ├── Dockerfile
│           ├── pyproject.toml (uv)
│           └── .venv/
```

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Benefits:

Slide 7: Container & Virtual Environment Options

Deployment Flexibility

Option 1: Dockerfile (Containerized)

```
FROM rocm/pytorch:latest
WORKDIR /app
COPY requirements.txt .
RUN pip install -r requirements.txt
COPY . .
CMD ["python", "inference.py"]
```

Option 2: uv + pyproject.toml (Modern Python)

```
[project]
name = "mobilesam-inference"
version = "1.0.0"
dependencies = [
    "torch>=2.0.0",
    "onnxruntime-rocm>=1.17.0",
]
```

Option 3: pip + requirements.txt (Traditional)

```
torch==2.0.0
onnxruntime-rocm==1.17.0
opencv-python==4.8.0
```

Model Distribution

Optimized Models Repository:

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Expanding Model Portfolio

Staying Current with Latest Open Source Models

Emerging Neural Alternatives:

- 🔄 **End-to-End Neural Pipelines** replacing traditional CV pipelines
  - Example: Neural SLAM replacing traditional SLAM
  - Example: Neural 3D reconstruction replacing classical methods
- 🆕 **Latest Model Categories to Track:**
  - Multimodal Foundation Models (GPT-4V alternatives)
  - Action-VLMs (RT-2, OpenVLA successors)
  - Neural Rendering (NeRF, Gaussian Splatting for robotics)
  - On-device Edge LLMs (< 3B parameters)

Strategy:

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Slide 9: Benchmarking & Documentation Strategy

GitHub Pages Integration

Documentation as Code:

```
docs/
├── index.md                    # Landing page
├── models/
│   ├── mobilesam.md           # Model-specific docs
│   ├── yolov12.md
│   └── benchmarks/
│       ├── robotics.md        # Benchmark results
│       └── industrial.md
├── guides/
│   ├── quick-start.md
│   └── deployment.md
└── _config.yml                # GitHub Pages config
```

Automated Workflow:

1. **Code + Benchmarks** → GitHub Repo

Slide 10: GitHub Pages Example - Mock Preview

Sample Documentation Site Layout

Homepage Preview:

Physical AI SDK Documentation

🚀 Quick Start | 📊 Benchmarks | 🤖 Models

📚 Model Catalog

```
├── Vision Models (12)
│   └── MobileSAM, YoloV12, EfficientNet...
├── VLM Models (5)
│   └── Qwen3-VL, LLaMA3.2-Vision...
└── Audio Models (3)
    └── Whisper, XTTS...
```

📈 Performance Dashboard

Model	FPS	Latency	Acc
MobileSAM	240fps	4.2ms	94%
YoloV12	240fps	4.2ms	92%
Whisper	100tok/s	10ms	96%

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Slide 12: Practical Example - Model Lifecycle

Case Study: Qwen3-VL Integration

Enablement:

- ✅ Model weights download and conversion
- ✅ Basic ONNX-RT integration
- ✅ Simple inference test (single image)
- ✅ Docker environment setup

Profiling

- 📊 Layer-wise performance analysis (AMD Profiler)
- 📊 Memory usage tracking
- 📊 Hotspot identification (Conv3D, GEMM operations)

Week 5-6: Benchmarking

- 🎯 Compute the relvent performance on a dataset wrt to aleaderboard or paper
- 🎯 KPI testing: Target 30fps @ 1080p
- 🎯 Accuracy validation on VQA datasets
- 🎯 Power consumption measurement

Fine-tuning:

- ⚙️ kernel optimization for GEMM
- ⚙️ INT8 quantization (W8A8)

Resource Management Strategy

Balancing Internal & External Resources

Resource Allocation Matrix:

Lifecycle Stage	Internal Team (PAVS)	External Partners	Duration
Enablement	▉▉▉▉▉▉▉▉▉▉ (Primary)	▒▒▒ (Support)	2-3 weeks
Profiling	▉▉▉▉▉▉▉▉ (Lead)	▉▉▉▉ (Assist)	2-4 weeks
Benchmarking	▉▉▉▉▉ (Guide)	▉▉▉▉▉▉▉ (Execute)	3-4 weeks
Fine-tuning	▉▉▉ (Review)	▉▉▉▉▉▉▉▉▉▉ (Primary)	4-8 weeks

Handoff Criteria:

✓ Model runs successfully on target hardware

✓ Profiling report completed

✓ Baseline benchmarks established

✓ Optimization opportunities documented

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Slide 14: Technology Watchlist 2026

Emerging Models & Technologies

High Priority:

1. **Qwen3-VL 32B** - Latest multimodal foundation model
2. **RT-X** - Robotics transformer with broader action space
3. **Neural SLAM** - End-to-end learned SLAM systems
4. **Depth Anything V2** - Improved monocular depth estimation

Medium Priority:

5. **SAM 2** - Segment Anything Model v2 with video support
6. **Florence-2** - Microsoft's vision foundation model
7. **Stable Diffusion 3** - For synthetic data generation
8. **Phi-3 Vision** - Compact VLM (< 4B params)

Success Metrics & KPIs

Measuring Lifecycle Effectiveness

Model Lifecycle KPIs:

Metric	Target	Current	Trend
Average Time to Enablement	< 2 weeks	1 weeks	⬇️ Improving
Models in Profiling Stage	5-8 concurrent	6	➡️ Stable
Benchmark Pass Rate (1st try)	> 80%	75%	⬆️ Improving
Fine-tuning Speedup (avg)	> 2x	2.3x	⬆️ Good
CI/CD Pipeline Success Rate	> 95%	97%	✅ Excellent

Portfolio Metrics:

- Total Models Supported: **45+**
- Production-Ready Models: **28**
- Models in Development: **12**
- Experimental/Research: **5**

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Roadmap & Next Steps

2026 Execution Plan

Q1 2026 (Current):

- ✅ Establish lifecycle framework
- 🔄 Complete 8 models through full lifecycle
- 🔄 Deploy GitHub Pages documentation

Q2 2026:

- 📋 Expand to 15 production-ready models
- 📋 Implement automated benchmarking CI/CD
- 📋 Onboard 2 external partner teams (MCW/SOW)

Q3 2026:

- 📋 Multi-model orchestration (5 system pipelines)
- 📋 Add 10 emerging models from watchlist

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